

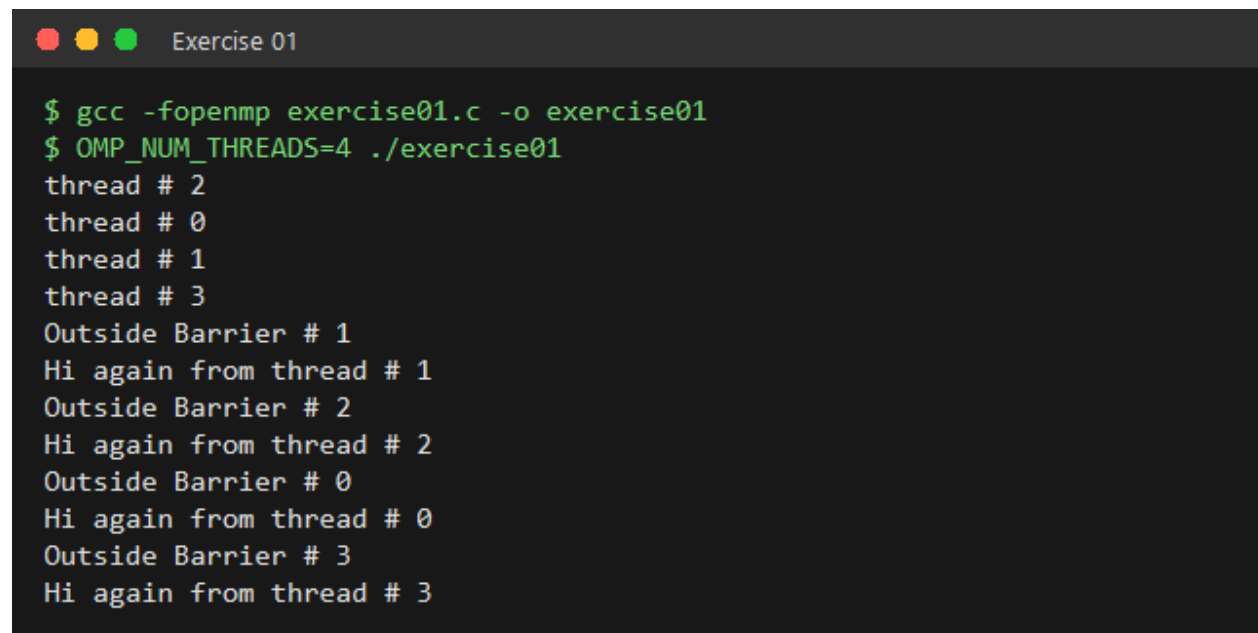
Exercise 1 - The #pragma omp barrier directive

Source file : exercise01.c

Build and execute

```
gcc -fopenmp exercise01.c -o exercise01
```

```
OMP_NUM_THREADS=4 ./exercise01
```



```
Exercise 01

$ gcc -fopenmp exercise01.c -o exercise01
$ OMP_NUM_THREADS=4 ./exercise01
thread # 2
thread # 0
thread # 1
thread # 3
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 2
Hi again from thread # 2
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 3
Hi again from thread # 3
```

What the barrier does in this program

- * The parallel region spawns a team of four threads and every one of them prints its own "thread # n" message.
- * The barrier directive acts as a meeting point: a thread that reaches it has to stop and wait until all the other members of the team have arrived as well.
- * As a result the four "thread # n" lines always appear first, and only then do the "Outside Barrier" / "Hi again" lines show up - the two groups of output are never interleaved.

* Within each group the ordering is still arbitrary, because once the threads are released they carry on running independently of each other.